

Stress states and point defect capture radii of dislocation a-loops and c-loops in alpha-zirconium

Jose F. March-Rico^{1,*}, Brian D. Wirth

Department of Nuclear Engineering, University of Tennessee, Knoxville, TN 37996, United States

HIGHLIGHTS

- Dislocation loop capture radii in zirconium were investigated with MS simulations.
- Spontaneous capture favors interstitial capture to a-loops and c-loops.
- Thermal drift capture introduces an inherent bias for same-type defect capture.
- Faulted basal pyramids (c-loop precursors) exhibit strong SIA bias for capture.

ARTICLE INFO

Keywords:

Zirconium
Modeling and simulation
c-loops and faulted basal pyramids
Multiscale modeling data
Defect capture radii to dislocation loops
SIA bias

ABSTRACT

Improving the predictive capability of microstructure evolution in irradiated α -zirconium requires a thorough understanding of the interaction rates between mobile defects and the existing, and evolving, microstructure. One of the parameters necessary to calculate these rates is the defect capture radii of dislocation loops. In order to better quantify this parameter, atomistic binding energy maps of point defects to interstitial a-loops, vacancy a-loops, and vacancy c-loops have been constructed using large-scale molecular statics simulations. Point defect capture radii have been correlated to the decay of binding energies with increasing distance using two criteria: spontaneous and thermal drift capture. Considering the spontaneous capture criteria, SIAs exhibit biased capture to all dislocation loops. However, when thermal drift capture is considered, the local stress state of dislocation loops imparts an inherent bias for the capture of same-type defects (i.e. vacancies to vacancy loops and vice versa). This phenomenon may drive the co-existence of vacancy and interstitial a-loops in irradiated α -Zr microstructure, an aspect that is poorly captured by most current models. The capture radii reported here will directly improve the predictive capability of microstructure evolution modeling in neutron-irradiated α -Zr.

1. Introduction

Materials exposed to neutron and ion irradiation will undergo significant alterations to their properties, depending on irradiation dose, dose rate and temperature, and may become deformed during their lifetime. These changes are often correlated to an evolution in the material microstructure. One of the challenges in predicting material performance under irradiation, then, is developing an accurate method to model microstructural evolution. Several models have been proposed for irradiated α -Zr with most recent attempts focused on cluster

dynamics (CD) modeling [1–4]. Improving the predictive capability of these models requires a thorough parameterization of defect production from displacement cascades [5–11], defect transport throughout the lattice [12–15], and the interaction rates between mobile defects with the existing, and evolving, microstructure.

Interaction rates are typically based on spherical reaction volumes for small clusters, and toroidal reaction volumes for dislocation loops [16,17]. A cluster can be considered to have a toroidal reaction volume when its outer radius notably exceeds its inner radius (which is equal to the capture radius for mobile defects). The defect capture radius affects

* Corresponding author.

E-mail address: jmarchri@alum.utk.edu (J.F. March-Rico).

¹ Present address: Naval Nuclear Laboratory, West Mifflin, PA, USA, 15122.

both the magnitude of the toroidal interaction rate constant, and the sharpness of the transition between spherical and toroidal reaction volumes [16]. Thus, it is imperative to calculate accurate values for the capture radii of defects in order to calculate dislocation loop capture rates.

Typically, the capture radius of a dislocation loop is assumed to be some multiple of the loop Burgers vector [2]. This method does not, however, differentiate between the capture tendencies of interstitial and vacancy a-loops in α -Zr, which share the same Burgers vector. As such, the differences in capture rates depends entirely on the diffusivity and concentration of the mobile species. In a CD framework, it would then be difficult to model the co-existence of interstitial and vacancy a-loops, as both of these loops would experience a net flux of either vacancies or interstitials. It is for this reason that vacancy a-loops are reported to not grow [1] or are simply ignored [3,4,18] in most models of microstructure evolution in α -Zr. Single interstitials and monovacancies are both considered to migrate with a preference for basal plane diffusion [12–15] and correspondingly, preferentially interact with prismatic a-loops [19]. Molecular dynamics simulations indicate that small interstitial clusters tend to form basal-plane configurations and migrate almost exclusively in the basal plane while small vacancy clusters migrate quasi-isotropically or with a preference for c-axis diffusion [12, 14]. Theoretically, this should result in a net flux of interstitials to vacancy a-loops and mitigate their growth. Despite this, vacancy a-loops account for at least half of visible microstructure in irradiated α -Zr, with sizes comparable to or larger than interstitial a-loops [20–25]. Although, it is important to note that definitively quantifying interstitial versus vacancy character of dislocation loops can be experimentally challenging [26].

One possibility to explain this behavior stems from a difference in the defect capture rates of interstitial and vacancy a-loops. Modeling the capture of defects by dislocation loops is a computationally expensive task requiring millions of atoms, and as such can only be performed with molecular statics (MS) simulations. To our knowledge, this has yet to be attempted in literature for dislocation loops in α -Zr. A similar work has, however, been performed to investigate the capture of point defects to small cavities in α -Fe [27]. It was found that the effective capture radii between voids and SIAs were notably larger than between voids and vacancies, resulting in biased capture of mobile SIA defects. This bias was found to persist for over-pressurized bubbles, but the magnitude of the bias was greatly reduced due to the compressive strain field acting against SIA capture. The inherent bias for small cavities could have a significant effect on vacancy cluster growth [28,29], although most models neglect any differences in the capture distances for SIA and vacancy defects [30–33].

In this work, we perform MS calculations with the BMD19.1 potential [34] to quantify the capture distances of point defects to interstitial a-loops, vacancy a-loops, and basal c-loops. Defect capture is calculated radial and normal to the loop face. Any correlations between defect capture behavior and the stress states of the loops in these two directions will be discussed, and the potential for biased capture of defects will be assessed.

2. Computational method

MD simulations were performed as implemented in the Large-Scale Atomic /Molecular Massively parallel Simulator (LAMMPS) code [35]. The BMD19.1 interatomic Zr-Zr potential [34] was selected due to its improved predictions of defect properties compared to the M07 [36] and A95 [37] potentials. Unlike previous Zr-Zr potentials, the BMD19.1 potential accurately predicts the basal-octahedral site as the preferred configuration for single SIAs [38–40], has close agreement with defect formation energies calculated by *ab-initio* methods [34], and accurately predicts that vacancy loops are thermodynamically preferred to spherical cavities in α -Zr [12]. The largest drawback of this potential is its severe underprediction of the basal stacking fault energies (SFE) [12,

34]. All energy minimizations were performed with the Polak-Ribiere version of the conjugate gradient algorithm with a relative energy tolerance of 10^{-12} and a force tolerance of 10^{-12} (eV/Å). Periodic boundary conditions were implemented for all simulations. The USER-OMP and USER-INTEL acceleration packages were integrated into LAMMPS to minimize computational time. All atomic positions were observed in the OVITO software [41]. Dislocation loop structures were visualized with common neighbor analysis (CNA) [42].

Interstitial and vacancy a-loops are known to form on, or near, $\{10\bar{1}0\}$ prismatic plane I (PPI) habit planes with perfect $\vec{b} = 1/3(11\bar{2}0)$ Burgers vector at large sizes [20,22,23,25,43,44]; vacancy c-loops instead form on (0001) basal planes with a Burgers vector of either $\vec{b} = 1/2[0001]$ (corresponding to basal extrinsic faulting) or $\vec{b} = 1/6(20\bar{2}3)$ (corresponding to basal I1 intrinsic faulting) [25,45–47]. For this work, interstitial loops were formed by inserting parallel basal-crowdions in a disk of radius R on a PPI habit plane (i.e. the lowest energy configuration identified by MS calculations with this potential [12]). A unit cell of this structure is visualized in Fig. 1 of Ref. [12]. Energy-minimizations were performed before and after dislocation loops were inserted into the simulation box. Due to the significant strain field imposed by SIA defects, these loops did not require annealing to reach their equilibrium configurations. To construct vacancy a-loops, atoms were removed from a double “corrugated” layer in a disk of radius R on a PPI habit plane (which is lower in energy than a single-layer prismatic structure [12, 48]). Basal c-loops, on the other hand, are lowest in energy in single-layer configurations [12] and, as such, were constructed by removing atoms from a disk of radius R from a single (0001) basal plane. Following the removal of atoms, the box was first energy-minimized and then ramped up to 573 K over 30 ps using an NPT (constant atom, pressure, and temperature) ensemble. The defect clusters were held at temperature for 200 ps before decreasing the temperature to 0 K over 30 ps and then performing a final energy minimization to calculate system energetics. Dislocation loop radii of 0.65, 1.2, 2.0, 3.0, and 5.0 nm were considered; this corresponds to approximate cluster sizes of 19, 55, 151, 333, and 955 defects. The simulation box size was set proportional to the loop radius: $8R \times 8R \times 8R$ (ranging from $\sim 40,000$ atoms for $R = 0.65$ nm to ~ 2.8 million atoms for $R = 5.0$ nm). This large simulation cell size was selected to minimize periodic image effects on the system energy. For example, the formation energy of a 939-interstitial dislocation loop is 374.63 eV, 377.47 eV, and 377.40 eV when using simulation box sizes of $6R \times 6R \times 6R$, $7R \times 7R \times 7R$, and $8R \times 8R \times 8R$, respectively; the loop energy is well-converged for a box scaling with $8R$.

For basal vacancy clusters containing fewer than 91 vacancies, basal extrinsic-faulted pyramids are the lowest energy configuration predicted by the BMD19.1 potential [12] and are currently hypothesized to act as stable nuclei for faulted c-loops [49,50]. These pyramids are characterized by an extrinsic-faulted base in the basal plane with pyramidal-faulted sides [49]. The BMD19.1 predicts that basal clusters between a size of 91 and 240 vacancies form basal extrinsic-faulted c-loops while basal clusters with greater than 240 vacancies form basal intrinsic-faulted c-loops [12]. Thus basal clusters designated with loop radii of 2.0, 3.0, and 5.0 nm are fully-formed dislocation c-loops after the NPT anneal; basal clusters designated with loop radii of 0.65 and 1.2 nm in fact form faulted basal pyramids after the NPT anneal, and this “loop radius” corresponds to the approximate radius of the faulted pyramidal base in the basal plane.

The binding energies of point defects to the minimized supercells of annealed defect structures were calculated by placing a single vacancy or SIA in the vicinity of the defect and performing an energy minimization. Single interstitials were placed in basal octahedral positions, as this is the preferred configuration predicted by the BMD19 potential and recent DFT studies [34,38–40]. Once minimized, the binding energies were calculated by Eqs. (1) and (2) for vacancies and interstitials, respectively:

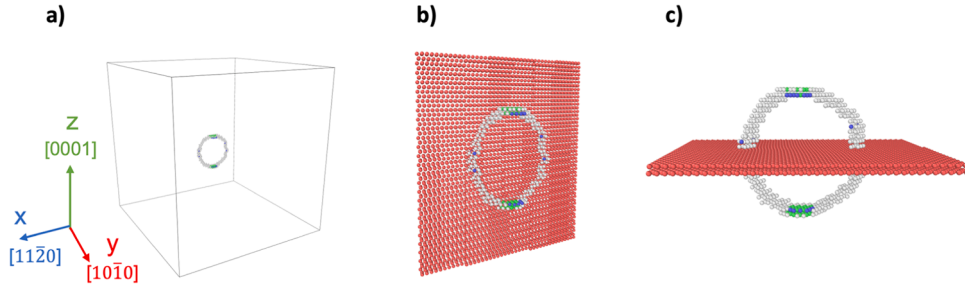


Fig. 1. (a) Simulation cell containing an interstitial a-loop visualized through CNA analysis (gray atoms are “unknown” structure, green atoms are FCC crystal structure). Cross-sections of atoms are taken through the (b) the face of the loop (radial capture) or (c) perpendicular to the loop.

$$E_v^b = E_{v,bulk}^f - [(E(N-1) + E_{Zr}^{coh}) - E(N)] \quad (1)$$

$$E_i^b = E_{i,bulk}^f - [E(N+1) - (E(N) + E_{Zr}^{coh})] \quad (2)$$

where $E(N)$, $E(N-1)$, and $E(N+1)$ are the total system energies for a system with N , $N-1$, or $N+1$ atoms, $E_{v,bulk}^f$ and $E_{i,bulk}^f$ are the bulk formation energies of a single interstitial (2.94 eV) and vacancy (2.03 eV), and E_{Zr}^{coh} is the cohesive energy of a single Zr atom (6.432 eV at 0 K) as calculated with the BMD19.1 potential. The positions of atoms in and around the defect clusters were used to determine insertion coordinates for vacancies and interstitials. Sampling the entire volume around the larger loops, however, would require a significant number of minimizations for each loop. Therefore, a cross-sectional region containing the face of the loop was considered. The region was three atom layers thick in the direction normal to the loop face and spanned twice its diameter in the directions parallel to the loop face to determine properties as a function of radial distance from the loop center (Fig. 1(b)). A similar cross-sectional slice rotated 90° such that it intersects the loop center (Fig. 1(c)) was considered to analyze properties perpendicular to the loop face. The radial and perpendicular interaction volumes for defect capture are visually illustrated in Fig. 2. It is worth noting again that the 0.65 nm and 1.2 nm basal clusters form three-dimensional faulted basal pyramids rather than dislocation loops; in these cases, the base of the pyramid is treated analogous to a loop face for this analysis.

Once atom positions radial or perpendicular to the loop were determined via the cross-sectional selection, average values for the local atomic stress as well as point defect binding energies could be determined. The stress tensors for each atom were calculated in LAMMPS in units of pressure*volume and converted into GPa by dividing the atomic volume. For properties radial to the loop face, average values were calculated by binning atoms into 0.15 nm elliptical bins that radiated from the loop center. The bins were given ellipticity factors that matched the ellipticity of the current loop being analyzed. As such, properties are averaged as a function of radial distance along the major axis from the loop center. For perpendicular properties, “line scans” were performed with 0.25 nm perpendicular bins along the direction normal to the loop face. The line scan analysis was performed at the center and periphery of the loop. The point defect capture radii could then be calculated in one of two ways. The spontaneous capture radius is defined as the distance from the loop at which the point defect binding energy falls to one-half of its maximum value [27]. The thermal drift capture radius is defined as the distance from the loop at which the binding energy falls below $\frac{3}{2}k_b T$ (and can no longer thermally escape drift towards the sink) [27].

3. Stress states surrounding dislocation loops in α -Zr

Atomic stresses surrounding dislocation loops are presented in average hydrostatic stress colormaps in Fig. 3, for which tensile and compressive stresses are positive (blue) and negative (red), respectively. Vacancy a-loops exhibit tensile stresses along their loop face, but this behavior reverses beyond the loop periphery; a weak compressive stress

is exerted radially on the surrounding bulk. Normal to the loop face, a long-range tensile stress extends along the primary a-axis direction that intersects the loop. TEM investigation of irradiated Zr shows that vacancy a-loops tend to order on parallel basal sheets with increasing fluence [22,51,52]. However, within these basal sheets, the orientations of vacancy a-loops are seemingly random [22]. This is qualitatively consistent with the stress state analysis given here; vacancy a-loops should re-orient such that their tensile stress fields do not overlap.

Interstitial a-loops (Fig. 3(c,d)) share similar stress state trends, but with an opposing sign relative to vacancy a-loops. These loops exhibit compressive stress along their faces but exert a radial tensile stress on the surrounding matrix; a compressive stress field extends normal to the loop face. Because interstitial and vacancy a-loops grow simultaneously in neutron-irradiated α -Zr [22,23,25,53], their complementary stress states are likely one of the many factors resulting in irradiation growth strain saturation at moderate fluence [54,55]. Basal c-loops (Fig. 3(e,f)), on the other hand, share similar stress behavior with vacancy a-loops, but with a 90° rotation. Weak compressive stresses are exerted radially from the c-loop periphery (along the a directions) while a long-range tensile stress is exerted normal to the loop face (along the c direction). This is consistent with breakaway irradiation growth behavior which results in a-axis expansion and c-axis contraction of single crystalline Zr [54,55].

In order to assess the magnitude of the atomic stresses, atoms were binned into 0.15 nm radial bins from the center of the loop for vacancy a-loops, interstitial a-loops, and vacancy c-loops (Fig. 4(a,c,e)), and into 0.25 nm perpendicular bins from the periphery of the loop to calculate averaged stress states (Fig. 4(b,d,f)). Considering the stresses plotted in the radial direction, the stresses near the loop center are tensile for vacancy loops and compressive for interstitial loops. The sign of this stress reverses beyond the loop periphery. Interestingly, the magnitude of the stress within the loop center decreases with increasing loop size. Presumably, this is due to the restoration of the original hcp lattice structure near the center of the dislocation loop. N de Diego et al. have shown for dislocation loops in α -Zr that the interatomic spacing at the loop center

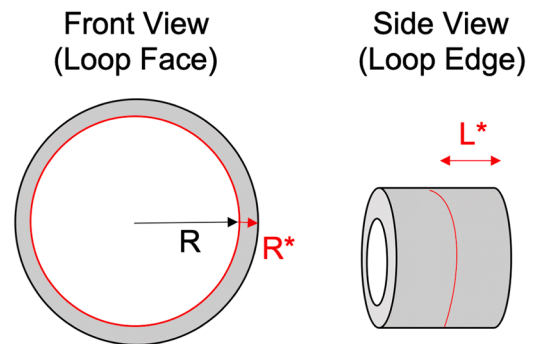


Fig. 2. The radial capture radius, R^* , defines the interaction volume radial from the periphery of the loop. The perpendicular capture distance, L^* , defines the interaction volume normal to the loop face.

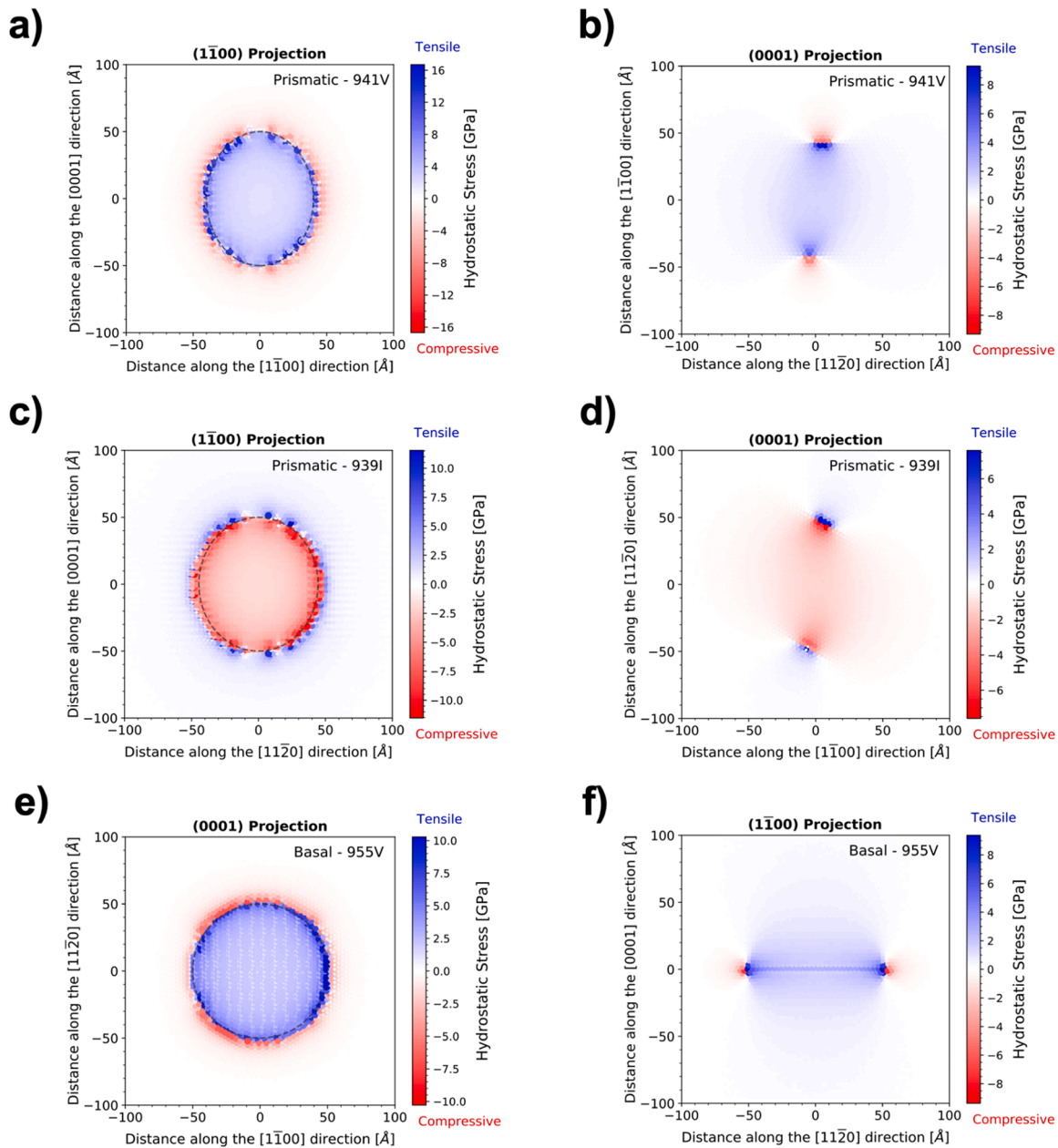


Fig. 3. Hydrostatic stress color maps for vacancy a-loops (a,b), interstitial a-loops (c,d), and vacancy c-loops (e,f). $(11\bar{2}0)$ projections in the first column demonstrate stress radial to the loop center while (0001) projections in the second column demonstrate stresses normal to the loop face.

is smaller than the bulk α -Zr spacing for interstitial loops (compressed) and larger than the bulk α -Zr spacing for vacancy loops (extended) [56]. As the loop size increases, the interatomic spacing at the center of the loop approaches the value of the bulk lattice. This behavior is consistent with the reduced atomic stress near the center of larger dislocation loops in Fig. 4(a,c,e). Lattice distortion is significant near the dislocation core at the periphery of the loop; as the loop radius widens and the dislocation core segments separate further, the interatomic spacing in the loop interior is able to relax. The structure relaxation with increasing loop size is the likely driver for the decreasing normal stresses seen in Fig. 4 (b,d,f).

4. Calculating point defect capture radii

4.1. Binding energy maps

Point defect absorption at sinks such as dislocation loops is driven by either a reduction in the local formation enthalpy of the defect, or the removal of the defect from the system. It has been shown that the formation energy per defect for dislocation loops decreases with their size (consider Fig. 5 in Ref. [12]). For example, the formation energy per vacancy in a 941 V prismatic a-loop is ~ 0.4 eV while the bulk formation energy for a single vacancy is predicted to be 2.03 eV by the BMD19.1

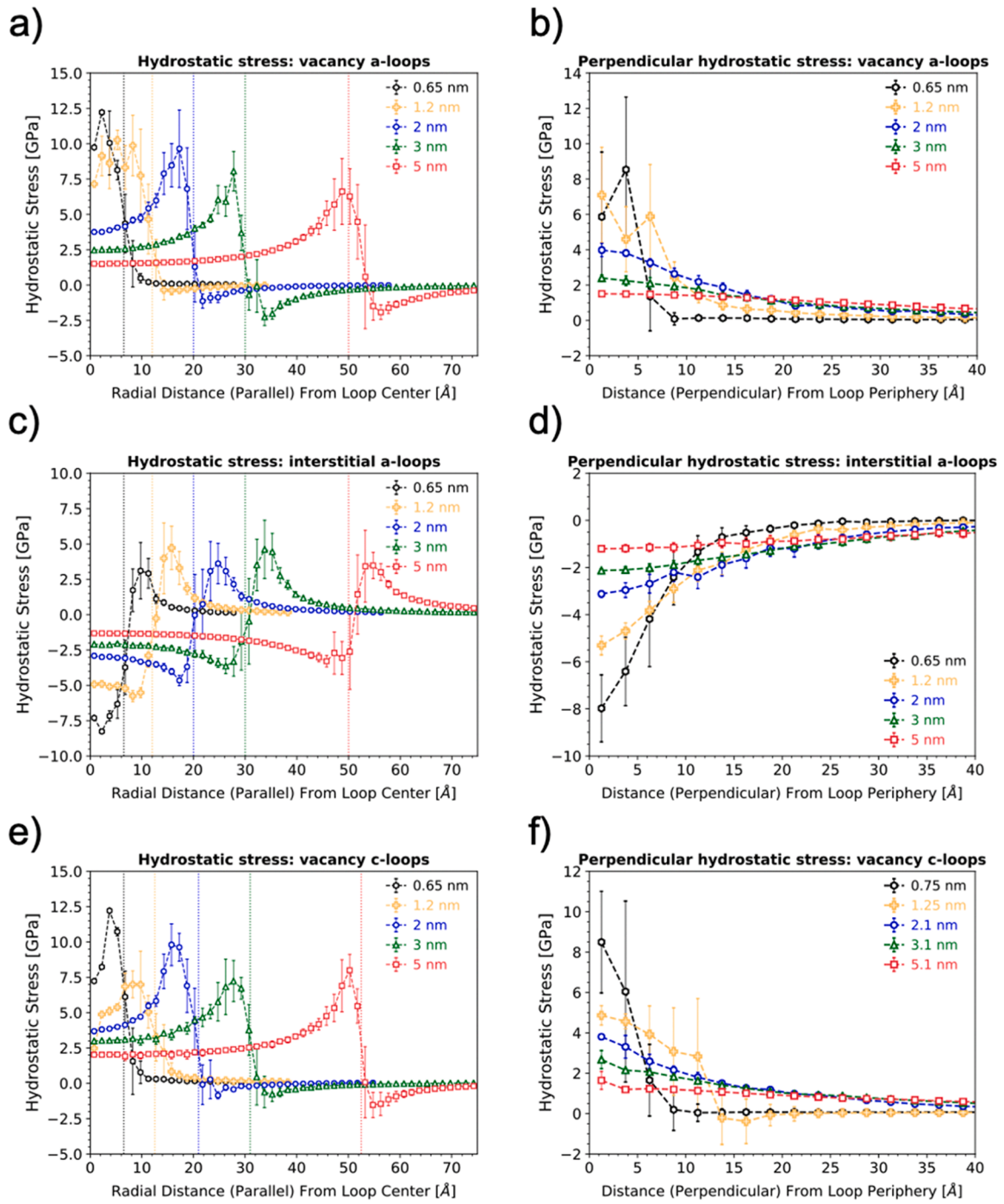


Fig. 4. Average values for the hydrostatic atomic stress plotted in 0.15 nm radial bins (a, c, e) and 0.25 nm perpendicular bins (b, d, f) for vacancy a-loops (first row), interstitial a-loops (middle row), and vacancy c-loops (bottom row). Positive stresses are tensile. Dotted lines indicate the position of the loop periphery (a,c,e).

potential [34]. Thus, there is a significant potential energy well for a migrating bulk vacancy to be captured by the vacancy loop. The same argument could be made for a single interstitial being captured by interstitial a-loops. Likewise, there is a significant enthalpic driving force for loops to capture and annihilate defects of the opposing type (i. e. a single interstitial annihilating through capture by a vacancy loop).

When considering defect capture by dislocation loops in a CD modeling framework, large dislocation loops are characterized by toroidal reaction volumes [16,17,57]. Functional solutions for the sink strength relating to toroidal reaction volumes are only valid when the loop outer radius notably exceeds the inner radius [58]. Thus, the reaction rates for defect capture at loops must be smoothly transitioned

between spherical and toroidal capture volumes, with a mixed regime at intermediate loop sizes. The necessity for this treatment is made clear with the binding energy maps for single SIA capture to vacancy (V) loops with sizes $n = 19$ V, 151 V, and 941 V in Fig. 5. It is clear that the binding energy is peaked near the periphery of these loops (i.e. there is some capture volume extending from the periphery of the loop). For loops with sizes $n = 19$ V, 151 V, and 941 V, the total reaction volume for a single SIA can be approximated as spherical (the capture volume fully overlaps), mixed (the capture volume partially overlaps), and toroidal (there is negligible overlap of the capture volume), respectively. There is an atomic mis-match between the dislocation loop and the surrounding matrix at the loop periphery; at the center of these loops, the hcp crystal

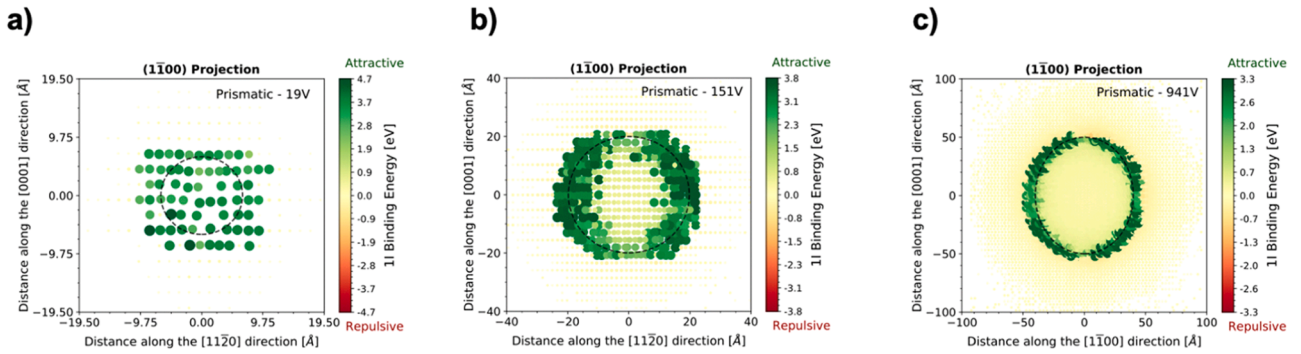


Fig. 5. Single interstitial binding energy maps to vacancy loops with a) 19 vacancies ($R = 0.65$ nm), b) 151 vacancies ($R = 2$ nm), and c) 941 vacancies ($R = 5$ nm).

structure is restored, although it is characterized by a slightly compressed lattice parameter for interstitial loops and a slightly extended lattice parameter for vacancy loops [12,56,59].

Monovacancy capture energies to vacancy a-loops, interstitial a-loops, and basal c-loops are plotted as a function of radial distance from the center of the loop in Fig. 6(a, c, e); corresponding single interstitial capture energies are plotted in Fig. 6(b, d, f). The locations of the loop peripheries are indicated with vertical dotted lines. Due to the dislocation loop structure, point defect binding energies are maximized at the loop periphery and minimal near the loop center (where the hcp structure has been restored, although slightly distorted). Point defect binding energies remain at an appreciable level for some distance beyond the periphery of the loop before decaying towards zero with increasing distance. This persistent binding energy beyond the loop periphery is related to the capture volume for point defects, and characteristic capture radii will be calculated in the following sections based on the rate of interaction decay with increasing distance.

Consider first the capture of the same type of defect to a dislocation loop (i.e. an interstitial to an interstitial loop in Fig. 6(d)). The general trend is that, as the loop size increases, the capture energy for the defect increases. This is consistent with the thermodynamic argument that there exists a deeper potential energy well for capture at larger loops. The opposite is true for defect annihilation at dislocation loops (Fig. 6(b, c, f)). The annihilation of an interstitial at a vacancy loop can be thought of as the recombination of a Frenkel pair. In this case, the capture energy is large due to the large formation energy of the bulk SIA, but the capture energy decreases with loop size because the annihilating vacancy has a lower effective formation energy. Due to the disparity in bulk formation energies for single SIAs and vacancies, single SIAs always observe higher capture energies towards vacancy loops. As such, interstitial loops are biased towards vacancy capture at small sizes but this behavior reverses for larger loop sizes. It is interesting to note the particularly low binding energy of single vacancies to small (< 1.2 nm) basal vacancy clusters (Fig. 6(e)), which form pyramidal configurations rather than dislocation c-loop configurations.

The capture energies of point defects perpendicular (rather than radial) to the loop periphery are plotted in Fig. 7. Appreciable binding is shown by single SIAs to vacancy a-loops and c-loops up to distances as large as ~ 10 Å (Fig. 7(b)). Similar, albeit shorter-range, capture behavior is seen for single vacancies. This implies that long-range interactions with point defects is most likely mediated by the stress field gradient, and that interstitials are more susceptible to local alterations of the stress field.

While the BMD19.1 appears to be a robust potential for the simulation of irradiation-induced defects, it is important to again note the considerable underprediction of basal stacking fault energies [12,34] in the context of the binding energies reported here. The total SFE energy is related to the size of the loop or pyramid, and the addition or removal of a defect from one of these basal clusters will result in a change in the total SFE. This change would normally be reflected in a reduction of the

binding energy (since the emission of a monomer would slightly decrease the total SFE). As such, the binding energies reported for basal clusters in Fig. 6(e,f) and Fig. 7(e,f) are expected to be slightly over-predicted by the BMD19.1 potential. Because the SFE effect on binding energies would systematically decrease their magnitudes, however, this would only result in a downward shift in the reported binding curves and is not expected to affect capture radius calculations.

4.2. Spontaneous capture radii of point defects

The spontaneous capture distance represents the separation distance at which a substantial drop in the interaction between the point defect and the dislocation loop has occurred. This is considered to occur when the binding energy falls to a threshold of half of its maximum value, as suggested by Kohnert et al. [27]. While this capture criteria is not as physically-based as the thermal drift criteria (Section 4.3), it is still considered for comparison. Table 1 provides the tabulated spontaneous capture distances for point defect capture. Single SIAs have greater spontaneous radial capture distances than monovacancies to vacancy loops of all sizes. The potential energy well is deeper for SIAs, and therefore the gradient is steeper, which drives longer-range capture of SIAs than vacancies. The radial capture radii, R^* , of SIAs to vacancy loops demonstrates that the depth of the potential energy well dominates relative to the stress state when considering spontaneous point defect capture. Similar to the preferential SIA capture by vacancy loops, monovacancies are captured at greater distances by small interstitial loops. Conversely, large interstitial loops have a greater capture radius for single SIAs than monovacancies. This can be understood by considering that larger SIA loops exhibit reduced formation energies per SIA, and thus offer a lower-energy state for migrating SIAs.

A comparison of the radial and perpendicular capture radii in Table 1 show that the capture distances are not symmetrical about the dislocation loop core. The capture distances perpendicular to the loop periphery are consistently larger than the radial capture distances. However, the same trends for preferential SIA or vacancy capture remain when considering these perpendicular radii. Incorporating these values would increase the effective capture volume of dislocation loops, particularly for small loops for which these capture distances notably exceed the cluster radius.

4.3. Thermal drift capture radii of point defects

The thermal drift analysis method is based on the concept that an interaction gradient between a defect and any given sink promotes defect drift diffusion towards that sink [27]. This effective capture radius is considered to be the distance at which the interaction energy falls below $\frac{3}{2}k_bT$. At lower temperatures, it becomes increasingly difficult for a defect to thermally migrate against the interaction gradient, and the capture distance becomes much greater than those calculated with the spontaneous method, as can be seen in a comparison of Table 1

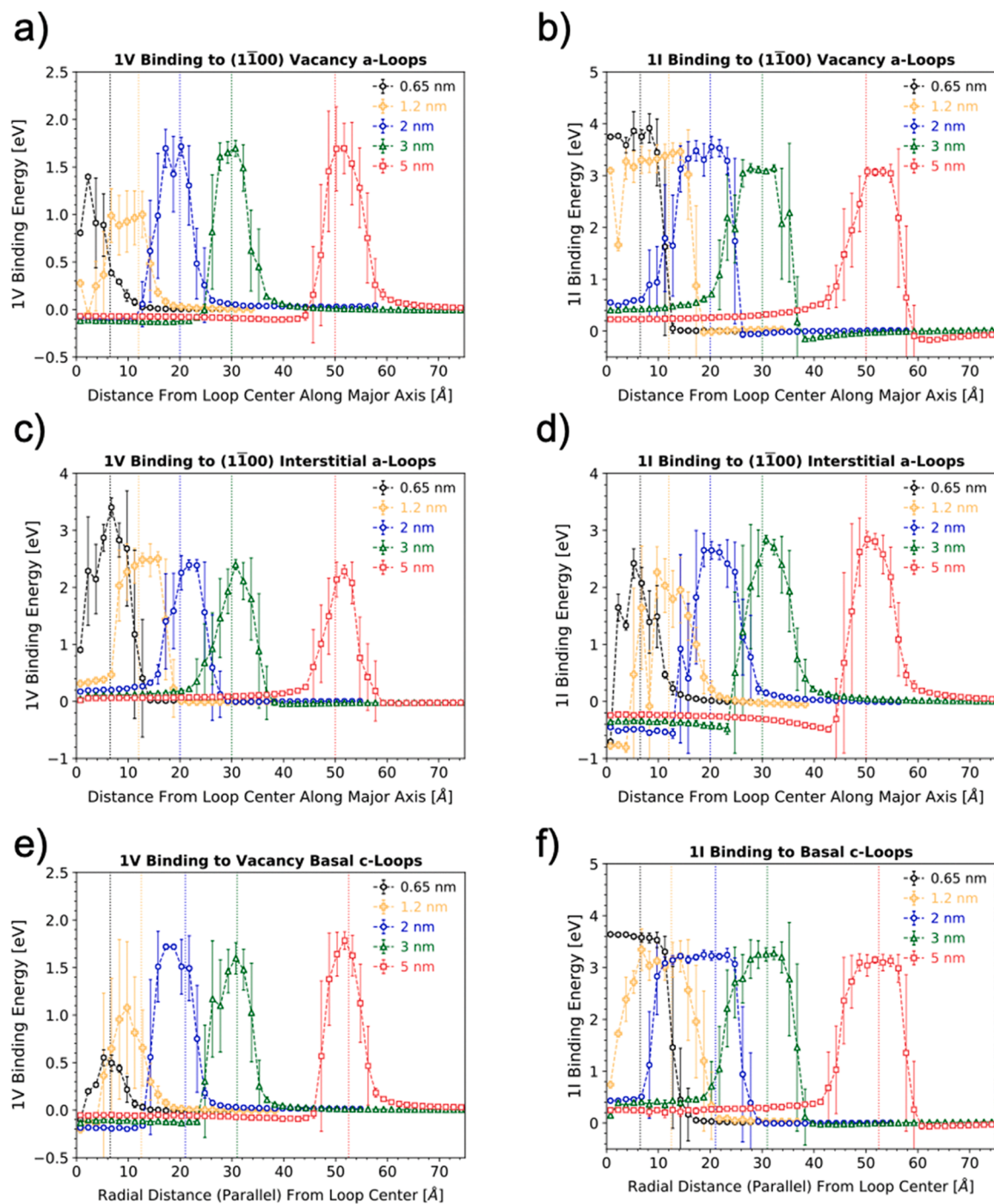


Fig. 6. Average values, in 0.15 nm radial bins, for the binding energies of single vacancies (a, c, e) and single interstitials (b, d, f) for (a,b) vacancy a-loops (c,d) interstitial a-loops and (e,f) vacancy c-loops. Positive binding energies are attractive. Dotted lines indicate the position of the loop periphery (a,c,e). Note that the clusters designated as “0.65 nm” and “1.2 nm” in (e,f) have faulted basal pyramid configurations.

(spontaneous) and Table 2 (thermal drift analyzed at 573 K).

When using the spontaneous capture analysis, SIAs are preferentially captured by vacancy a-loops of all sizes; however, using the thermal drift criteria, monovacancies now exhibit greater capture distances to large vacancy a-loops. Likewise, single SIAs are captured at greater distances than monovacancies at interstitial a-loops when using the thermal drift criteria. These observations appear correlated with the radial stress states exerted by vacancy and interstitial a-loops, respectively. Vacancy a-loops exert a short-range compressive stress on the surrounding matrix in the directions radial to the loop periphery (Fig. 3(a), Fig. 4(a)), and

the magnitude of the compressive stress increases with loop size. This is qualitatively consistent with the preferential absorption of vacancies at vacancy a-loops. In contrast, interstitials are preferentially absorbed at interstitial a-loops, and this is consistent with the short-range tensile stresses exerted radial to the loop.

These observations can be rationalized through the consideration of elastic dipole tensors for point defects [60–63]. The dipole components within the basal plane and along the c-axis have been computed for vacancies and various interstitial configurations based on *ab initio* and molecular statics calculations [60]. Dipole components, P_{ij} , for point

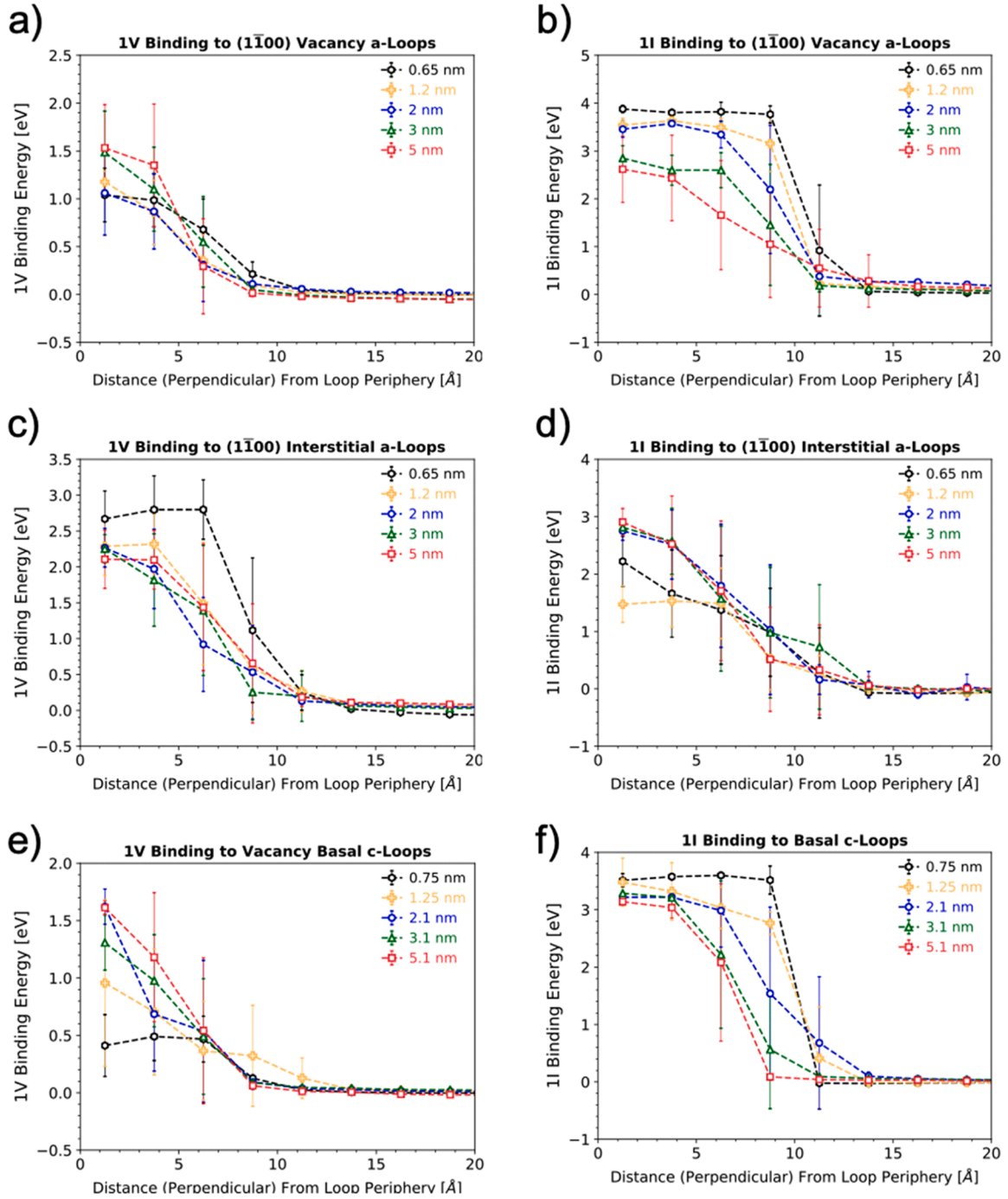


Fig. 7. Average values, in 0.25 nm perpendicular bins, for the binding energies of single vacancies (a, c, e) and single interstitials (b, d, f) for (a,b) vacancy a-loops (c, d) interstitial a-loops and (e,f) vacancy c-loops. Positive binding energies are attractive. Note that the clusters designated as “0.75 nm” and “1.25 nm” in (e,f) have faulted basal pyramid configurations.

defects in α -Zr were calculated with the BMD19.1 potential based on the average residual stress on the simulation box, $\langle\sigma_{ij}\rangle$, following atomic relaxation as outlined in Ref. [60]:

$$P_{ij} = -V\langle\sigma_{ij}\rangle \quad (3)$$

where V is the simulation box volume. The chosen coordinate system is $e_1 \parallel [2\bar{1}\bar{1}0]$, $e_2 \parallel [01\bar{1}0]$, and $e_3 \parallel [0001]$ such that the dipole tensors are diagonal; for this coordinate system, $P_{11} = P_{22}$ for a basal-octahedral SIA and monovacancy. The calculated dipole components in a simulation box containing 12,800 atoms ($20 \times 20 \times 16$ primitive hcp cells) are

reported in Table 3 for the BMD19.1 potential and compared with reported values based on *ab initio* calculations [60]. Table 3 indicates that the dipole values are in reasonably good agreement, which provides further confidence in the BMD19.1 potential in calculating defect interactions in α -Zr. Vacancy dipole components have the opposite sign and are lower in magnitude than those of self-interstitials. Because the interaction energy between point defects and an induced strain field depends on the sign and magnitude of the elastic dipole components [63], the preferential capture of point defects at loops of the same character (i.e. interstitials at interstitial loops) is better understood. The higher magnitude of interstitial dipole components indicates that

Table 1

Spontaneous defect capture distances in the directions radial to (R^*) and normal to (L^*) the loops. Radii are determined by the “half-max” criteria, indicating the distance at which the capture energy of the point defect drops off to one-half of the maximum value.

Loop Radius (Å)	Effective radial capture radius, R^* (spontaneous criteria) (Å)					
	Prismatic Vacancy a-loop		Prismatic Interstitial a-loop		Basal Vacancy c-loop	
	1I	1V	1I	1V	1I	1V
6.5	4.5	0	3.8	4.4	5.0	2.0
12	4.7	2.2	5.2	5.5	5.4	1.0
20	4.8	2.9	6.0	5.3	4.8	2.1
30	5.8	3.4	6.0	4.9	5.6	3.3
50	6.9	5.9	5.9	4.5	6.6	4.4

Loop Radius (Å)	Effective perpendicular capture radius, R^* (spontaneous criteria) (Å)					
	Prismatic Vacancy a-loop		Prismatic Interstitial a-loop		Basal Vacancy c-loop	
	1I	1V	1I	1V	1I	1V
6.5	10.4	7.1	7.9	8.3	10.0	7.9
12	9.9	5.1	8.1	7.2	9.8	5.4
20	9.3	5.3	7.6	5.7	8.6	3.4
30	8.8	5.4	6.9	6.8	7.1	5.4
50	7.7	5.1	6.8	7.5	6.9	5.2

Table 2

Thermal drift capture distances as determined by the $k_b T$ criteria in the directions radial (R^*) and normal (L^*) to the loops at $T = 573$ K.

Loop Radius (Å)	Effective radial capture radius, R^* (thermal drift criteria, 573 K) (Å)					
	Prismatic Vacancy a-loop		Prismatic Interstitial a-loop		Basal Vacancy c-loop	
	1I	1V	1I	1V	1I	1V
6.5	6.3	4.8	9.5	6.6	9.2	3.5
12	5.7	5.4	10.1	6.9	11.1	4.5
20	5.2	10.1	15.6	7.7	7.2	5.8
30	6.4	9.5	18.9	6.7	7.1	7.3
50	8.0	14.1	21.8	7.7	8.3	10.6

Loop Radius (Å)	Effective perpendicular capture radius, R^* (thermal drift criteria, 573 K) (Å)					
	Prismatic Vacancy a-loop		Prismatic Interstitial a-loop		Basal Vacancy c-loop	
	1I	1V	1I	1V	1I	1V
6.5	14.4	10.2	11.9	12.4	10.2	9.6
12	24.8	9.3	12.2	19.6	12.4	12.2
20	29.3	10.9	13.1	17.7	15.7	9.7
30	32.0	7.7	12.7	14.8	14.8	10.0
50	34.6	11.6	13.1	15.3	15.2	10.4

Table 3

Elastic dipole tensor components for point defects in α -Zr. The chosen coordinate system is $e_1 \parallel [2\bar{1}\bar{1}0]$, $e_2 \parallel [01\bar{1}0]$, and $e_3 \parallel [0001]$ such that the dipole tensors are diagonal. Dipole components were calculated via the residual stress method outlined in Ref. [60] with a supercell size of $N = 12,800$ atoms.

Calculation Method	Ref.	Monovacancy		Basal-octahedral SIA	
		P_{11}	P_{33}	P_{11}	P_{33}
BMD19	This work	-4.78	-4.68	17.0	13.0
<i>Ab initio</i>	[60]	-5.14	-7.62	17.0	10.6

interstitials should exhibit a higher sensitivity to induced strain fields than vacancies. This is evident in a comparison of spontaneous and thermal drift capture radii in Tables 1 and 2. Thermal drift capture of interstitials to interstitial loops is much longer ranged than the corresponding spontaneous capture, and this relative difference is larger than

that observed for vacancy capture at vacancy loops. It is also interesting to note the point defect capture tendencies in Table 2 for basal c-loops. For small sizes where c-loop precursors take the form of faulted basal pyramids [12,49], single SIAs are preferentially captured over monovacancies. It is not until a faulted basal c-loop is fully formed on a single basal plane that vacancies observe preferential capture. Faulted basal pyramids exhibit strong normal tensile stresses and negligible radial compressive stresses (Fig. 4(e,f)); relative to basal pyramids, c-loops exhibit weaker normal tensile stresses and stronger radial compressive stresses. Considering the elastic dipole argument for point defects, the shift in preferential point defect capture tendencies between basal pyramids and fully-formed c-loops may be understood.

When using the thermal drift criteria for calculating capture radii, the resultant capture volumes are sensitive to the operating temperature. Fig. 8 demonstrates this temperature dependence for radial point defect capture at dislocation loops with radius $R = 5$ nm, which exhibits a decaying dependence on inverse temperature:

$$r_d^{\text{radial}} = \alpha_d \left(\frac{T_m}{T} \right)^{m_d} \quad (4)$$

where T_m is the melting temperature of α -Zr (2128 K). Physically, m_d describes the sensitivity of the thermal drift capture radii with temperature; larger values of m_d indicate a stronger sensitivity to temperature.

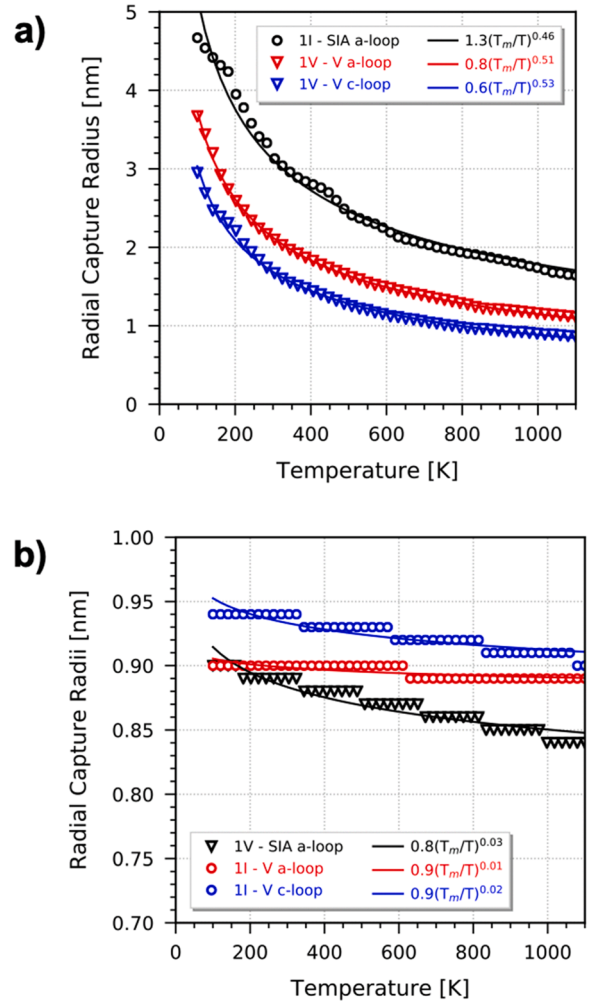


Fig. 8. Thermal drift capture radii for single SIA (1I) and monovacancy (1V) capture by SIA a-loops, vacancy a-loops, and vacancy c-loops of size $R = 5$ nm as a function of temperature. Capture radii are shown for the capture of (a) same-type defects and (b) different-type defects.

The α_d parameter is the magnitude of the thermal drift capture radii that the function asymptotically approaches at high temperature. It could be argued that the calculation of α_d is an alternative method to estimate the spontaneous capture radius; α_d represents the distance at which a defect would be captured irrespective of the system temperature (i.e. spontaneous capture). For the capture of same-type defects (i.e. single interstitial capture to interstitial a-loops) in Fig. 8(a), there is a notable temperature dependence. When instead considering the capture of opposing-type defects (i.e. monovacancy capture at interstitial a-loops) in Fig. 8(b), there is only a weak temperature dependence. This can again be understood based on the stress fields of the dislocation loops and the elastic dipole components of point defects [60]. For the case of interstitial a-loops, a tensile stress field is exerted radially from the loop periphery. Migrating single interstitials observe an attractive interaction gradient along this tensile stress field. Thus, temperature plays a greater role on the thermal drift capture of migrating interstitials to interstitial a-loops, and vacancies to vacancy loops. The necessary parameters (α_d , m_d) to calculate these temperature-dependent radii are provided in Table 4 for each loop type and size investigated in this work.

5. Net SIA biases to dislocation loops in α -Zr

For a given dislocation loop L , the bias for point defect capture can be written in terms of the loop sink strength for absorption of interstitials (i) and vacancies (v):

$$B_L = \frac{Z_{i,L}}{Z_{v,L}} - 1 \quad (5)$$

The calculation of $Z_{i,L}$ and $Z_{v,L}$ is dependent on the size of the loop. Large dislocation loops can be considered to have toroidal (T) reaction volumes, and an approximation for the sink strength depends on the defect capture radius, r_d [16,17]:

$$Z^T = \frac{4\pi^2 R_L \rho_L}{\ln(1 + 8R_L/(r_1 + r_d))} \quad (6)$$

where R_L is the loop radius, ρ_L is the loop density, and r_1 is the point defect radius. However, this approximation is only sufficient once the inner radius notably exceeds the loop radius. For small loops, the sink strength is best approximated by a spherical (S) reaction volume:

$$Z^S = 4\pi(r_i + r_j + r_d)\rho_L \quad (7)$$

Table 4

Necessary parameters to calculate thermal drift capture radii ($\text{\AA}$) as a function of temperature using Eq. (3). For each loop size and type, these parameters are given for a single SIA (1I) and single vacancy (1V) capture.

Loop Radius ($\text{\AA}$)	α_d Parameter ($\text{\AA}$)					
	Prismatic Vacancy a-loop		Prismatic Interstitial a-loop		Basal Vacancy c-loop	
	1I	1V	1I	1V	1I	1V
6.5	5.3	3.6	7.5	7.0	6.9	3.0
12	6.5	3.5	9.1	7.6	7.9	3.7
20	6.1	5.2	11.3	8.0	7.6	3.3
30	7.1	6.2	12.3	7.2	7.4	3.7
50	8.9	7.8	12.6	8.3	8.9	5.9
Loop Radius ($\text{\AA}$)	m_d Parameter (unitless)					
	Prismatic Vacancy a-loop		Prismatic Interstitial a-loop		Basal Vacancy c-loop	
	1I	1V	1I	1V	1I	1V
6.5	0.16	0.33	0.23	0.075	0.29	0.29
12	0.0095	0.47	0.25	0.032	0.19	0.28
20	0.0098	0.56	0.29	0.055	0.062	0.59
30	0.025	0.41	0.37	0.047	0.058	0.64
50	0.0070	0.51	0.46	0.032	0.019	0.53

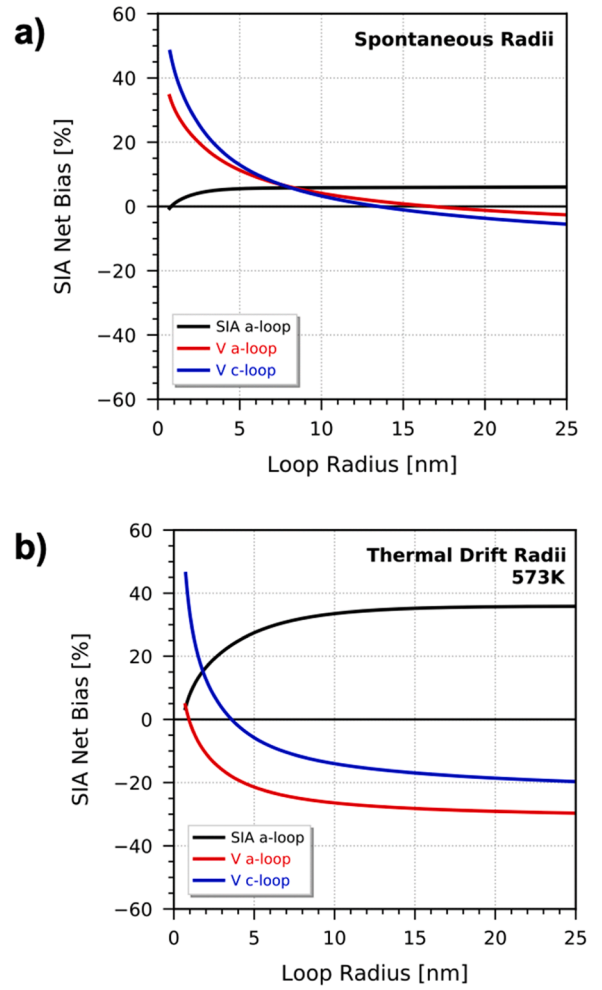


Fig. 9. Net SIA biases to SIA a-loops, vacancy a-loops, and vacancy c-loops using (a) spontaneous capture radii and (b) thermal drift capture radii calculated at 573 K.

Dislocation loop sink strengths are smoothly transitioned between spherical and toroidal reaction kinetics, such that the effective sink strength can be calculated by:

$$Z_L = (1 - \alpha_j)Z^T + \alpha_j Z^S \quad (8)$$

$$\alpha_j = \left(1 + \left(\frac{R_L}{3(r_1 + r_d)}\right)^2\right)^{-1} \quad (9)$$

Using the capture radii tabulated in Tables 1 and 2, the net SIA biases for dislocation loops of varying size are plotted in Fig. 9(a) for spontaneous capture and Fig. 9(b) for thermal drift capture. Note that because the SIA bias is a ratio of the interstitial and vacancy sink strengths for a given loop, the values shown in Fig. 9 are invariant on the dislocation loop density. The effect of the capture radii criterion is significant. When considering spontaneous capture, single SIAs generally have larger interaction volumes. There is a net positive SIA bias for all dislocation loops less than $R \sim 15$ nm. When considering thermal drift capture, there is an even stronger SIA bias for capture at SIA a-loops. For vacancy a-loops, however, there is now a strong bias for monovacancy, rather than SIA, capture. This phenomenon is quite notable and likely promotes simultaneous SIA and vacancy a-loop growth in irradiated α -Zr. It is common in reaction rate theory models to simply assume that SIAs have a ~ 10 – 20 % bias for capture at all dislocation loops (similar to what is demonstrated with spontaneous capture radii in Fig. 9(a)) [1,57]. When using this assumption, CD frameworks have been unable to predict the

simultaneous growth of interstitial and vacancy a-loops [1,3,4,18]. Thermal drift capture radii, which are influenced by the unique stress states for each dislocation loop, provide a physically-relevant and inherent bias for vacancy and interstitial a-loops to preferentially absorb vacancy versus SIA point defects (e.g., same type defects), respectively.

However, should be highlighted that these simulations of point defect capture to dislocation loops were performed in a pristine simulation box of α -Zr at zero pressure. In a true irradiated material, the local stress state may be altered by factors such as the loop concentration and spatial arrangement, and this may modify the thermal drift capture distances. The spontaneous capture criteria, however, is relatively insensitive to alterations in external factors such as temperature and stress state. As such, the thermal drift capture distances reported here likely act as an upper bound on the interaction distance while the spontaneous capture distances act as a lower bound.

It is interesting to note that small c-loops are biased towards SIA capture while large c-loops are biased towards vacancy capture. Preferential SIA absorption by nucleating c-loops could explain why these defects observe delayed growth; such that once these defects have grown to a size where vacancy biases become relevant, their growth is quite rapid. It is crucial to note that the net biases in Fig. 9 do not take defect diffusive anisotropy into account, and are derived only from the discrepancy of interaction volumes for point defects. An accurate representation of dislocation interaction rates must also incorporate the anisotropy of defect diffusion [12,19]. For the case presented here, the toroidal sink strength (Eq. (5)) would need to be modified by the anisotropic capture efficiency, which will be different for a-loops (which have PPI habit planes) and c-loops (which have basal habit planes). Because single SIAs and monovacancies migrate with similar anisotropy [12], however, it is not expected to qualitatively affect the relative biases shown in Fig. 9.

6. Summary and conclusions

The stress states and capture radii of point defects to dislocation loops in α -Zr were investigated with the BMD19.1 potential. Vacancy loops exert short-range radial compressive stresses and long-range tensile normal stresses; the magnitude of the radial compressive stress generally increases with loop size. The same trends are true for interstitial loops, but with opposite sign of the stress state. The spontaneous capture distances of point defects are predominantly related to the depth of the potential energy well, which depends on the magnitude of the defect formation energy in the bulk and its maximum capture energy to the dislocation loops. When analyzing spontaneous defect capture, SIAs observe biased capture to all dislocation loop types. However, when using the thermal drift analysis method, however, it was found that point defect capture biases were correlated with the stress states surrounding dislocation loops. Monovacancies exhibited biased radial capture to vacancy a-loops while single SIAs observe biased radial capture to interstitial a-loops. This discrepancy in the interaction volumes of point defects provides a physically-relevant and inherent bias that may help explain the co-existence of interstitial and vacancy a-loops in irradiated α -Zr. It is anticipated that the point defect capture radii reported here will improve the predictive capability of cluster dynamics models.

CRedit authorship contribution statement

Jose F. March-Rico: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Brian D. Wirth:** Conceptualization, Formal analysis, Resources, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

This research was performed under appointment to the Rickover Fellowship Program in Nuclear Engineering sponsored by the Naval Reactors Division of the U.S. Department of Energy. The authors gratefully acknowledge Dr. Li Yang (UTK) for her discussion and collaboration on the calculation of the dipole tensors with the BMD19 interatomic potential.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

- [1] F. Christien, A. Barbu, Cluster dynamics modelling of irradiation growth of zirconium single crystals, *J. Nucl. Mater.* 393 (1) (2009) 153–161, <https://doi.org/10.1016/j.jnucmat.2009.05.016>.
- [2] F. Christien, A. Barbu, Effect of self-interstitial diffusion anisotropy in electron-irradiated zirconium: a cluster dynamics modeling, *J. Nucl. Mater.* 346 (2–3) (2005) 272–281, <https://doi.org/10.1016/j.jnucmat.2005.06.024>.
- [3] S.I. Choi, G.G. Lee, J. Kwon, J.H. Kim, Modeling of sink-induced irradiation growth of single-crystal and polycrystal zirconiums in nuclear reactors, *J. Nucl. Mater.* 468 (2016) 56–70, <https://doi.org/10.1016/j.jnucmat.2015.11.014>.
- [4] Y. Li, N. Ghoniem, Cluster dynamics modeling of irradiation growth in single crystal Zr, *J. Nucl. Mater.* 540 (2020), 152312, <https://doi.org/10.1016/j.jnucmat.2020.152312>.
- [5] R.E. Voskoboinikov, Y.N. Osetsky, D.J. Bacon, Identification and morphology of point defect clusters created in displacement cascades in α -zirconium, *Nucl. Instrum. Methods Phys. Res. Sect. B* 242 (1–2) (2006) 530–533, <https://doi.org/10.1016/j.nimb.2005.08.167>.
- [6] W. Zhou, J. Tian, Q. Feng, J. Zheng, X. Liu, J. Xue, D. Qian, S. Peng, Molecular dynamics simulations of high-energy displacement cascades in hcp-Zr, *J. Nucl. Mater.* 508 (2018) 540–545, <https://doi.org/10.1016/j.jnucmat.2018.06.002>.
- [7] Q.U.A. Sahi, Y.S. Kim, Molecular dynamics simulations of the coupled effects of quench and temperature on displacement cascades in α -zirconium, *Nucl. Eng. Technol.* 50 (6) (2018) 907–914, <https://doi.org/10.1016/j.net.2018.04.013>.
- [8] S.J. Wooding, D.J. Bacon, A molecular dynamics study of displacement cascades in α -zirconium, *Philos. Mag. A* 76 (5) (1997) 1033–1051, <https://doi.org/10.1080/01418619708200013>.
- [9] S.J. Wooding, L.M. Howe, F. Gao, A.F. Calder, D.J. Bacon, A molecular dynamics study of high-energy displacement cascades in α -zirconium, *J. Nucl. Mater.* 254 (2–3) (1998) 191–204, [https://doi.org/10.1016/S0022-3115\(97\)00365-6](https://doi.org/10.1016/S0022-3115(97)00365-6).
- [10] J.F. March-Rico, C.M. Mcswain, B.D. Wirth, Quantifying the impact of an electronic drag force on defect production from high-energy displacement cascades in α -zirconium, *J. Nucl. Mater.* 542 (2020), 152539, <https://doi.org/10.1016/j.jnucmat.2020.152539>.
- [11] F. Gao, D.J. Bacon, L.M. Howe, C.B. So, Temperature-dependence of defect creation and clustering by displacement cascades in α -zirconium, *J. Nucl. Mater.* 294 (3) (2001) 288–298, [https://doi.org/10.1016/S0022-3115\(01\)00483-4](https://doi.org/10.1016/S0022-3115(01)00483-4).
- [12] J.F. March-Rico, B.D. Wirth, Defect cluster configurations and mobilities in α -Zr: a comparison of the BMD19 and M07 interatomic potentials, *J. Nucl. Mater.* 559 (2022), 153441, <https://doi.org/10.1016/j.jnucmat.2021.153441>.
- [13] N. De Diego, Y.N. Osetsky, D.J. Bacon, Mobility of interstitial clusters in alpha-zirconium, *33(13)* (2002) 783–789, 10.1007/s11661-002-1008-2.
- [14] N. De Diego, A. Serra, D.J. Bacon, Y.N. Osetsky, On the structure and mobility of point defect clusters in alpha-zirconium: a comparison for two interatomic potential models, *Modell. Simul. Mater. Sci. Eng.* 19 (3) (2011), 035003, <https://doi.org/10.1088/0965-0393/19/3/035003>.
- [15] G.D. Samolyuk, A.V. Barashev, S.I. Golubov, Y.N. Osetsky, R.E. Stoller, Analysis of the anisotropy of point defect diffusion in hcp Zr, *Acta Mater.* 78 (2014) 173–180, <https://doi.org/10.1016/j.actamat.2014.06.024>.
- [16] A. Kohnert, B. Wirth, Cluster dynamics models of irradiation damage accumulation in ferritic iron. II. Effects of reaction dimensionality, *J. Appl. Phys.* 117 (2015), 154306, <https://doi.org/10.1063/1.4918316>.
- [17] A. Seeger, U. Gösele, Steady-state diffusion of point defects to dislocation loops, *Phys. Lett. A* 61 (6) (1977) 423–425, [https://doi.org/10.1016/0375-9601\(77\)90355-3](https://doi.org/10.1016/0375-9601(77)90355-3).
- [18] B. Christiaen, C. Domain, L. Thuinet, A. Ambard, A. Legris, Influence of vacancy diffusional anisotropy: understanding the growth of zirconium alloys under

- irradiation and their microstructure evolution, *Acta Mater.* 195 (2020) 631–644, <https://doi.org/10.1016/j.actamat.2020.06.004>.
- [19] C.H. Woo, Theory of irradiation deformation in non-cubic metals: effects of anisotropic diffusion, *J. Nucl. Mater.* 159 (1988) 237–256, [https://doi.org/10.1016/0022-3115\(88\)90096-7](https://doi.org/10.1016/0022-3115(88)90096-7).
- [20] D. Northwood, R.W. Gilbert, L.E. Bahen, P.M. Kelly, R.G. Blake, A. Jostsons, P. K. Madden, D. Faulkner, W. Bell, R.B. Adamson, Characterization of neutron irradiation damage in zirconium alloys — an international “round-robin” experiment, *J. Nucl. Mater.* 79 (1979) 379–394, [https://doi.org/10.1016/0022-3115\(79\)90103-X](https://doi.org/10.1016/0022-3115(79)90103-X).
- [21] P.M. Kelly, R.G. Blake, A. Jostsons, An interpretation of corduroy contrast in neutron irradiated zirconium, *J. Nucl. Mater.* 59 (3) (1976) 307–315, [https://doi.org/10.1016/0022-3115\(76\)90063-5](https://doi.org/10.1016/0022-3115(76)90063-5).
- [22] A. Jostsons, P.M. Kelly, R.G. Blake, The nature of dislocation loops in neutron irradiated zirconium, *J. Nucl. Mater.* 66 (3) (1977) 236–256, [https://doi.org/10.1016/0022-3115\(77\)90113-1](https://doi.org/10.1016/0022-3115(77)90113-1).
- [23] A. Jostsons, et al., Neutron Irradiation-Induced Defect Structures in Zirconium, ASTM International, 1979, pp. 61–1979, <https://doi.org/10.1520/STP38157S>.
- [24] M. Griffiths, Microstructure Evolution in Zr Alloys During Irradiation: Dose, Dose Rate, and Impurity Dependence, ASTM International, 2010, pp. 436–2010, <https://doi.org/10.1520/JAI101146>.
- [25] M. Griffiths, A review of microstructure evolution in zirconium alloys during irradiation, *J. Nucl. Mater.* 159 (1988) 190–218, [https://doi.org/10.1016/0022-3115\(88\)90093-1](https://doi.org/10.1016/0022-3115(88)90093-1).
- [26] M. Kirk, X. Yi, M. Jenkins, Characterization of irradiation defect structures and densities by transmission electron microscopy, *J. Mater. Res.* 30 (2015) 1195–1201, <https://doi.org/10.1557/jmr.2015.19>.
- [27] A.A. Kohnert, M.A. Cusentino, B.D. Wirth, Molecular statics calculations of the biases and point defect capture volumes of small cavities, *J. Nucl. Mater.* 499 (2018) 480–489, <https://doi.org/10.1016/j.jnucmat.2017.12.005>.
- [28] T. Okita, W.G. Wolfer, A critical test of the classical rate theory for void swelling, *J. Nucl. Mater.* 327 (2) (2004) 130–139, <https://doi.org/10.1016/j.jnucmat.2004.01.026>.
- [29] M.F. Wehner, W.G. Wolfer, Vacancy cluster evolution in metals under irradiation, *Philos. Mag.* A 52 (2) (1985) 189–205, <https://doi.org/10.1080/01418618508237618>.
- [30] H. Trinkaus, B.N. Singh, S.I. Golubov, Progress in modelling the microstructural evolution in metals under cascade damage conditions, *J. Nucl. Mater.* 283–287 (2000) 89–98, [https://doi.org/10.1016/S0022-3115\(00\)00332-9](https://doi.org/10.1016/S0022-3115(00)00332-9).
- [31] A. Barbu *, C.S. Becquart, J.L. Bocquet, J. Dalla Torre, C. Domain, Comparison between three complementary approaches to simulate ‘large’ fluence irradiation: application to electron irradiation of thin foils, *Philos. Mag.* 85 (4–7) (2005) 541–547, <https://doi.org/10.1080/14786430412331334616>.
- [32] C. Domain, C.S. Becquart, L. Malerba, Simulation of radiation damage in Fe alloys: an object kinetic Monte Carlo approach, *J. Nucl. Mater.* 335 (1) (2004) 121–145, <https://doi.org/10.1016/j.jnucmat.2004.07.037>.
- [33] C.J. Ortiz, M.J. Caturla, Cascade damage evolution: rate theory versus kinetic Monte Carlo simulations, *J. Comput. Aided Mater. Des.* 14 (1) (2007) 171–181, <https://doi.org/10.1007/s10820-007-9082-9>.
- [34] E. Wimmer, M. Christensen, W. Wolf, W.H. Howland, B. Kammenzind, R.W. Smith, Hydrogen in zirconium: atomistic simulations of diffusion and interaction with defects using a new embedded atom method potential, *J. Nucl. Mater.* 532 (2020), 152055, <https://doi.org/10.1016/j.jnucmat.2020.152055>.
- [35] S. Plimpton, Fast parallel algorithms for short-range molecular dynamics, *J. Comput. Phys.* 117 (1) (1995) 1–19, <https://doi.org/10.1006/jcph.1995.1039>.
- [36] M.I. Mendelev, G.J. Ackland, Development of an interatomic potential for the simulation of phase transformations in zirconium, *Philos. Mag. Lett.* 87 (5) (2007) 349–359, <https://doi.org/10.1080/09500830701191393>.
- [37] G.J. Ackland, S.J. Wooding, D.J. Bacon, Defect, surface and displacement-threshold properties of α -zirconium simulated with a many-body potential, *Philos. Mag.* A 71 (3) (1995) 553–565, <https://doi.org/10.1080/01418619508244468>.
- [38] G.D. Samolyuk, S.I. Golubov, Y.N. Osetsky, R.E. Stoller, Self-interstitial configurations in hcp Zr: a first principles analysis, *Philos. Mag. Lett.* 93 (2) (2013) 93–100, <https://doi.org/10.1080/09500839.2012.745653>.
- [39] G. V  rit  , C. Domain, C.C. Fu, P. Gasca, A. Legris, F. Willaime, Self-interstitial defects in hexagonal close packed metals revisited: evidence for low-symmetry configurations in Ti, Zr, and Hf, *Phys. Rev. B* 87 (13) (2013), 134108, <https://doi.org/10.1103/PhysRevB.87.134108>.
- [40] C. Varvenne, F. Bruneval, M.C. Marinica, E. Clouet, Point defect modeling in materials: coupling *ab initio* and elasticity approaches, *Phys. Rev. B* 88 (13) (2013), 134102, <https://doi.org/10.1103/PhysRevB.88.134102>.
- [41] A. Stukowski, Visualization and analysis of atomistic simulation data with OVITO—the open visualization tool, *Modell. Simul. Mater. Sci. Eng.* 18 (1) (2009), 015012, <https://doi.org/10.1088/0965-0393/18/1/015012>.
- [42] D. Faken, H. J  nsson, Systematic analysis of local atomic structure combined with 3D computer graphics, *Comput. Mater. Sci.* 2 (2) (1994) 279–286, [https://doi.org/10.1016/0927-0256\(94\)90109-0](https://doi.org/10.1016/0927-0256(94)90109-0).
- [43] M. Griffiths, R.W. Gilbert, V. Fidleris, R.P. Tucker, R.B. Adamson, Neutron damage in zirconium alloys irradiated at 644 to 710 k, *J. Nucl. Mater.* 150 (2) (1987) 159–168, [https://doi.org/10.1016/0022-3115\(87\)90071-7](https://doi.org/10.1016/0022-3115(87)90071-7).
- [44] M. Griffiths, M.H. Loretto, R.E. Smallman, Electron damage in zirconium, *J. Nucl. Mater.* 115 (2–3) (1983) 323–330, [https://doi.org/10.1016/0022-3115\(83\)90323-9](https://doi.org/10.1016/0022-3115(83)90323-9).
- [45] C. Varvenne, O. Mackain, E. Clouet, Vacancy clustering in zirconium: an atomic-scale study, *Acta Mater.* 78 (2014) 65–77, <https://doi.org/10.1016/j.actamat.2014.06.012>.
- [46] M. Griffiths, R.W. Gilbert, The formation of c-component defects in zirconium alloys during neutron irradiation, *J. Nucl. Mater.* 150 (2) (1987) 169–181, [https://doi.org/10.1016/0022-3115\(87\)90072-9](https://doi.org/10.1016/0022-3115(87)90072-9).
- [47] M. Griffiths, R.A. Holt, A. Rogerson, Microstructural aspects of accelerated deformation of Zircaloy nuclear reactor components during service, *J. Nucl. Mater.* 225 (1995) 245–258, [https://doi.org/10.1016/0022-3115\(94\)00687-3](https://doi.org/10.1016/0022-3115(94)00687-3).
- [48] V.G. Kapinos, Y.N. Osetsky, P.A. Platonov, Computer simulation of vacancy loops and stacking faults in zirconium, *J. Nucl. Mater.* 195 (1) (1992) 83–101, [https://doi.org/10.1016/0022-3115\(92\)90365-R](https://doi.org/10.1016/0022-3115(92)90365-R).
- [49] B. Christiaen, C. Domain, L. Thuinet, A. Ambard, A. Legris, A new scenario for (c) vacancy loop formation in zirconium based on atomic-scale modeling, *Acta Mater.* 179 (2019) 93–106, <https://doi.org/10.1016/j.actamat.2019.07.030>.
- [50] C. Dai, P. Saidi, M. Topping, L.K. B  land, Z. Yao, M.R. Daymond, A mechanism for basal vacancy loop formation in zirconium, *Scr. Mater.* 172 (2019) 72–76, <https://doi.org/10.1016/j.scriptamat.2019.07.006>.
- [51] A. Harte, D. J  dern  s, M. Topping, P. Frankel, C.P. Race, J. Romero, L. Hallstadius, E.C. Darby, M. Preuss, The effect of matrix chemistry on dislocation evolution in an irradiated Zr alloy, *Acta Mater.* 130 (2017) 69–82, <https://doi.org/10.1016/j.actamat.2017.03.024>.
- [52] E. Francis, R.P. Babu, A. Harte, T.L. Martin, P. Frankel, D. J  dern  s, J. Romero, L. Hallstadius, P.A.J. Bagot, M.P. Moody, M. Preuss, Effect of Nb and Fe on damage evolution in a Zr-alloy during proton and neutron irradiation, *Acta Mater.* 165 (2019) 603–614, <https://doi.org/10.1016/j.actamat.2018.12.021>.
- [53] A. Jostsons, R.G. Blake, J.G. Napier, P.M. Kelly, K. Farrell, Faulted loops in neutron-irradiated zirconium, *J. Nucl. Mater.* 68 (3) (1977) 267–276, [https://doi.org/10.1016/0022-3115\(77\)90251-3](https://doi.org/10.1016/0022-3115(77)90251-3).
- [54] G.J.C. Carpenter, R.A. Murgatroyd, A. Rogerson, J.F. Watters, Irradiation growth of zirconium single crystals, *J. Nucl. Mater.* 101 (1) (1981) 28–37, [https://doi.org/10.1016/0022-3115\(81\)90441-4](https://doi.org/10.1016/0022-3115(81)90441-4).
- [55] G.J.C. Carpenter, R.H. Zee, A. Rogerson, Irradiation growth of zirconium single crystals: a review, *J. Nucl. Mater.* 159 (1988) 86–100, [https://doi.org/10.1016/0022-3115\(88\)90087-6](https://doi.org/10.1016/0022-3115(88)90087-6).
- [56] N. De Diego, Y.N. Osetsky, D.J. Bacon, Structure and properties of vacancy and interstitial clusters in α -zirconium, *J. Nucl. Mater.* 374 (1–2) (2008) 87–94, <https://doi.org/10.1016/j.jnucmat.2007.07.011>.
- [57] A.A. Kohnert, B.D. Wirth, L. Capolungo, Modeling microstructural evolution in irradiated materials with cluster dynamics methods: a review, *Comput. Mater. Sci.* 149 (2018) 442–459, <https://doi.org/10.1016/j.commatsci.2018.02.049>.
- [58] V. Jansson, L. Malerba, d.b. Andree, C. Becquart, C. Domain, Sink strength calculations of dislocations and loops using OKMC, *J. Nucl. Mater.* 442 (2013) 218–226, <https://doi.org/10.1016/j.jnucmat.2013.08.052>.
- [59] Y.N. Osetsky, D.J. Bacon, A. Serra, Thermally activated glide of small dislocation loops in metals, *Philos. Mag. Lett.* 79 (5) (1999) 273–282, <https://doi.org/10.1080/095008399177345>.
- [60] C. Varvenne, E. Clouet, Elastic dipoles of point defects from atomistic simulations, *Phys. Rev. B* 96 (2017), 224103, <https://doi.org/10.1103/PhysRevB.96.224103>.
- [61] V.M. Chernov, D.A. Chulkin, A.B. Sivak, Interaction of intrinsic point defects with dislocation stress fields in hcp zirconium crystal, *Crystallogr. Rep.* 55 (2010) 83–96, <https://doi.org/10.1134/S10663774510010141>.
- [62] C. Varvenne, F. Bruneval, M.C. Marinica, E. Clouet, Point defect modeling in materials: coupling *ab initio* and elasticity approaches, *Phys. Rev. B* 88 (2013), 134102, <https://doi.org/10.1103/PhysRevB.88.134102>.
- [63] E. Clouet, C. Varvenne, T. Jourdan, Elastic modeling of point-defects and their interaction, *Comput. Mater. Sci.* 147 (2018) 49–63, <https://doi.org/10.1016/j.commatsci.2018.01.053>.